Type MS · Write a short number program from a list of requirements

Answer 1 of 6 · Question 3 part whole · 6 marks

2022 JUN · Paper 02 · extract 1 of 6 · find part whole in this mark scheme extract

3	**Award 1 mark for each of:**			
	A1	At least one variable with a meaningful name (1)		
	A2	At least one condition and price correct (ignore order) (1)		
	A3	All conditions and prices correct (1)		
	A4	Calculation for Total cost is correct (1)		
	A5	Price per textbook and total cost displayed (1)		
	A6	Calculation of cost and display of output are not included in the if statement (1)		(6)

Code examples

Python

```
# Q03

#  Initialise variables
quantity = 0
price = 0
cost = 0

#  Print prompt and take number of textbooks required
quantity = int(input("Please enter the quantity of books required: "))

#  Generate and display the price per textbook and the total cost
if quantity >= 1 and quantity <= 5:
    price = 20
elif quantity < 10:
    price = 15
else:
    price = 12

cost = price * quantity
print("The price per book is",price)
print("The total cost is",cost)
```

Type MS · Write a short number program from a list of requirements

Answer 1 of 6 · Question 3 part whole · 6 marks

2022 JUN · Paper 02 · extract 2 of 6 · find part whole in this mark scheme extract

Question	mp	Answe	Additional Guidance	Mark

Type MS · Write a short number program from a list of requirements

Answer 1 of 6 · Question 3 part whole · 6 marks

2022 JUN · Paper 02 · extract 3 of 6 · find part whole in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
6	A1	At least two appropriate subprograms used		(1)
	A2	Repeat guess until the word has been guessed **OR** no attempts left		(1)
	A3	Generate and display the number of attempts left	Must see working correctly	(1)
	A4	Request input of the word		(1)
	A5	Validate the input of the word to ensure it is the same length as the random word		(1)
	A6	Check to see if the guess matches the random word		(1)
	A7	If the word has been guessed, display message including the random word and number of attempts		(1)
	A8	Keep track and display letters that are in the word	Must see working	(1)
	A9	Keep track and display letters that are not in the word	Must see working	(1)
	A10	Letters appear only once in each list	Must see working	(1)
	A11	Display lose message including the random word		(1)

Band 1 (1-3 marks)	Band 2 (4-6 marks)	Band 3 (7-9 marks)	Mark
Little attempt to decompose into component parts	Some attempt to decompose into component parts	The problem has been decomposed into component parts	
Some parts of the logic are clear and appropriate to the problem	Most parts of the logic are clear and mostly appropriate to the problem	The logic is clear and appropriate to the problem	
Some appropriate use and manipulation of data types, variables, data structures and program constructs	The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate	The use and manipulation of data types, variables and data structures and program constructs is appropriate	
Parts of the code are clear and readable	Code is mostly clear and readable	Code is clear and readable	
Finished program will not be flexible enough with other data sets or input	Finished program will function with some but not all other data sets or input	Finished program could be used with other data sets or input	
The program meets some of the given requirements	The program meets most of the given requirements	The program fully meets the given requirements	(9)

Type MS · Write a short number program from a list of requirements

Answer 1 of 6 · Question 3 part whole · 6 marks

2022 JUN · Paper 02 · extract 4 of 6 · find part whole in this mark scheme extract

Code examples

Python

Subprograms

```
# Add your subprograms here
def checkInput(pWordLength):
    # Validate input to make sure it is the same length as the secret word
    check = ""
    while len(check) != pWordLength:
        check = input("Enter your guess. The secret word has " +str(pWordLength)+ " letters: ")
    return(check)

def checkLettersInWord(pGuess, pWordToGuess, pLetters, type):
    for letter in pGuess: # Check each letter in the guess
        if type == 0: # The check to carry out is for correct letters
            # If the letter is in the secret word and not already in the correctletters, add it
            if letter in pWordToGuess and letter not in pLetters:
                pLetters = pLetters + letter
        else: # The check to carry out is for wrong letters
            # If the letter is not in the secret word and not already in the wrongLetters, add
            if letter not in pWordToGuess and letter not in pLetters:
                pLetters = pLetters + letter
    return pLetters

def display(pCorrectLetters, pWrongLetters):
    # Display the correct and wrong letters
    if len(pCorrectLetters) > 0:
        print("Letter(s) in the secret word:",pCorrectLetters)

    if len(pWrongLetters) > 0:
        print("Letter(s) not in the secret word:",pWrongLetters)
```

Type MS · Write a short number program from a list of requirements

Answer 1 of 6 · Question 3 part whole · 6 marks

2022 JUN · Paper 02 · extract 5 of 6 · find part whole in this mark scheme extract

Main program

```
# Add your main program code here
wordLength = len(wordToGuess)
maxAttempts = wordLength + 3
numAttempts = 0
correctLetters = ""
wrongLetters = ""
guessed = False

# Loop until the no attempts are left or the secret word has been guessed
while numAttempts < maxAttempts and guessed == False:
    print("You have",maxAttempts - numAttempts,"attempts to guess the secret word.")
    numAttempts = numAttempts + 1

    guess = checkInput(wordLength)  # Validate the input

    if guess == wordToGuess: # Check to see if the secret word has been guessed
        guessed = True
        print("Well done. You guessed the secret word was",wordToGuess,"in",numAttempts,"attempts.")

    else: # If it hasn't been guessed check for correct and wrong letters
        correctLetters = checkLettersInWord(guess,wordToGuess, correctLetters, 0)
        wrongLetters = checkLettersInWord(guess,wordToGuess, wrongLetters, 1)
        display(correctLetters, wrongLetters)

# Display game over message if attempts have run out and the secret word has not been guessed
if numAttempts == maxAttempts and guessed == False:
    print("Game over. You did not guess that the secret word was",wordToGuess)
```

Type MS · Write a short number program from a list of requirements

Answer 1 of 6 · Question 3 part whole · 6 marks

2022 JUN · Paper 02 · extract 6 of 6 · find part whole in this mark scheme extract

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Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 1 of 17 · find part c in this mark scheme extract

Q3(b)	More memory/storage required (1) Because not sorted in place / not efficient / many steps / division of data set / is recursive (1)		(2)

Question	Answer	Additional Guidance	Mark

Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 2 of 17 · find part c in this mark scheme extract

Q3(a)	Award 1 mark each for: 1. File opened for input (1) 2. File opened for output (1) 3. Loop for processing data (1) 4. Check for '@' symbol (1) 5. Write invalid address to output file (1) 6. Close files (1)	Amended code should be saved as Q03aFINISHED. Do not penalise candidates who attempt more than the stated requirements.	(6)

Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 3 of 17 · find part c in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 4 of 17 · find part c in this mark scheme extract

Q3(c)	Award 1 mark for each section of code that: 1. Asks for input of a number and stores it in a variable with a meaningful name (1) 2. Checks number is between 1 and 50 (1) 3. Calculates square and cube of number (1) 4. Displays suitable labels e.g. 'Number', 'Square' and 'Cube' and relevant values (1) For the whole code: 5. Compiles and runs without logical or syntax errors (1) 6. Efficient in terms of computation, storage and selection of programming constructs (1)	Amended code should be saved as Q03cFINISHED. Allow follow through for mark point 4 when square and cube are incorrectly calculated. Credit alternative coding which produces same results. Accept use of exponent operators: C# e.g. Math.Pow (2, 3) = 8 Java e.g. Math.pow (2, 3) = 8 Python e.g. 2**3 = 8.	(6)

Question	Designated marks:	Additional Guidance	Mark

Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 5 of 17 · find part c in this mark scheme extract

Java Example code

Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 6 of 17 · find part c in this mark scheme extract

3(a)

```
package q03aFINISHED;

import java.io.BufferedReader;
import java.io.FileNotFoundException;
import java.io.FileReader;
import java.io.PrintWriter;
import java.io.UnsupportedEncodingException;
import java.util.Scanner;

public class Q03aFINISHED
{
    public static void main(String[] args) throws FileNotFoundException,UnsupportedEncodingException
    {
        String email="";
        PrintWriter errorFile;

        //  Open the input file

        Scanner emailFile = new Scanner(new BufferedReader(new FileReader("Email.txt")));

        //  Open the output file

        errorFile = new PrintWriter("Error.txt", "UTF-8");

        //  Find errors and write to output file

        while (emailFile.hasNextLine())
        {
            email = emailFile.nextLine();
            if (!email.contains("@"))
            {
                errorFile.println(email);
            }
        }

        //  Close files

        emailFile.close();
        errorFile.close();
    }
}
```

Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 7 of 17 · find part c in this mark scheme extract

3(c)	Note: Line 19 is an alternative response to line 18
	
	`package q03cFINISHED;`
	
	`import java.util.Scanner;`
	
	`public class Q03c`
	`{`
	`  public static void main(String[] args)`
	`  {`
	`  // Write your code here`
	`  Scanner input = new Scanner(System.in);`
	`  int myValue = 1;`
	`  while (myValue >= 1 && myValue <= 50)`
	`  {`
	`  System.out.println("Enter a value from 1 to 50: ");`
	`  myValue = input.nextInt();`
	`  if (myValue >= 1 && myValue <= 50)`
	`  {`
	`  System.out.println("Value " + myValue + " Square " + myValue * myValue + " Cube " + myValue * myValue * myValue);`
	`  System.out.println("Value " + myValue + " Square " + Math.pow(myValue, 2) + " Cube " + Math.pow(myValue, 3));`
	`  }`
	`  }`
	`  }`
	`}`	

Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 8 of 17 · find part c in this mark scheme extract

5

```
        int total = 0;
        double average;
        int books[]  = new int[libraryRecord.length];
        int gold = 0, silver = 0, bronze = 0;
        int goldID = 0, silverID = 0, bronzeID = 0;

        System.out.println("Pupil ID of pupils who have read less than 10 books");

            for (int i = 0; i < libraryRecord.length; i++)
            {
                books[i] = Integer.parseInt(libraryRecord[i][3]);
                if (books[i] > gold)
                {
                    bronze = silver; silver = gold; gold = books[i]; goldID = i;
                }
                else if (books[i] > silver)
                {
                    bronze = silver; silver = books[i]; silverID = i;
                }
                else if (books[i] > bronze)
                {
                    bronze = books[i]; bronzeID = i;
                }
                total = total + books[i];

                if (books[i] < 10)
                {
                    System.out.println(libraryRecord[i][0]);
                }
            }
            System.out.println();

            System.out.println("Total books read by pupils = " + total);

            System.out.println();

            average = total / libraryRecord.length;

            System.out.println("Average books read by pupils = " + average);

            System.out.println();

            System.out.println("Gold winner is " + libraryRecord[goldID][1] + " " + libraryRecord[goldID][2]);
            System.out.println("Silver winner is " + libraryRecord[silverID][1] + " " + libraryRecord[silverID][2]);
            System.out.println("Bronze winner is " + libraryRecord[bronzeID][1] + " " + libraryRecord[bronzeID][2]);
```

Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 9 of 17 · find part c in this mark scheme extract

Python Example code

1(c)

```
# Q01c

count = 0

maxValue = 5
constantValue = 7

while (count < maxValue):
    print(count, count + constantValue)
    count = count + 1
```

1(f)

```
    if (shopIncome > 5000) or (assistantSales >= 10):
        print ("Assistant bonus = ", assistantSalary * 0.1)

    elif (shopIncome >= 2000) and (assistantSales >= 5):
        print("Assistant bonus = ", assistantSalary * 0.05)
```

Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 10 of 17 · find part c in this mark scheme extract

2(a)

```
# Initialise variables
counter = 1
answer = randint(1,10)
guess = 0

# Print prompt and take guess from user
guess = int(input("Enter a number from 1 to 10: "))

# Create while loop to check answer
while guess!=answer:
    counter = counter + 1
    if(guess > answer):
        print (guess, " was too high. Try again.")
    else:
        print (guess, " was too low. Try again.")

    guess = int(input("Guess again: "))

# Report the correct answer to the user and indicate the number of guesses
print ("You guessed", guess, "in", counter, "guesses")
```

Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 11 of 17 · find part c in this mark scheme extract

3(a)

```
Q03aFINISHED

#   Open file and input data
with open('Email.txt','r') as inputFile:
    emailList = inputFile.readlines()

#   Open file for output
outputFile = open('Error.txt', 'a')

#   Find errors and write to error log
for address in emailList:
    if not "@" in address:
        outputFile.write(address)

#   Close files
outputFile.close()
inputFile.close()
```

Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 12 of 17 · find part c in this mark scheme extract

3(c)

```
1  # Q03c
2
3  # Write your code below this line
4  myValue = 1
5
6  while (myValue >= 1 and myValue <= 50):
7      myValue = int(input("Enter a value from 1 to 50: "))
8      if (myValue >= 1 and myValue <= 50):
9          print("Value: ", myValue, "Square: ",myValue*myValue, "Cube: ", myValue*myValue*
           myValue)
10
11         print("Value: ", myValue, "Square: ", myValue**2, "Cube: ", myValue**3)
```

Note: Line 11 is an alternative response to line 9.

Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 13 of 17 · find part c in this mark scheme extract

5

```
# Write your code below this line

total = 0
gold = 0
silver = 0
bronze = 0
goldID = 0
silverID = 0
bronzeID = 0

print ("Pupil ID of pupils who have read less than 10 books ")

for row in range(len(libraryRecord)):
    books = int(libraryRecord[row][3])
    if (books > gold):
        bronze = silver
        silver = gold
        gold = books
        goldID = row
    elif (books > silver):
        bronze = silver
        silver = books
        silverID = row
    elif (books > bronze):
        bronze = books
        bronzeID = row

    total = total + books

    if (books < 10):
        print ("\t",libraryRecord[row][0])

print ("\n")
print ("Total number of books read by pupils: " , total)

print ("\n")
print ("Average number of books read by pupils: " , total / len(libraryRecord[0]))

print ("\n")
print ("Gold medal winner is: \t\t", libraryRecord[goldID][1], " ", libraryRecord[goldID][2])
print ("Silver medal winner is: \t", libraryRecord[silverID][1], " ", libraryRecord[silverID][2])
print ("Bronze medal winner is: \t", libraryRecord[bronzeID][1], " ", libraryRecord[bronzeID][2])
```

Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 14 of 17 · find part c in this mark scheme extract

Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 15 of 17 · find part c in this mark scheme extract

Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 16 of 17 · find part c in this mark scheme extract

Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 17 of 17 · find part c in this mark scheme extract

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Type MS · Write a short number program from a list of requirements

Answer 3 of 6 · Question 2 part b · 7 marks

2020 NOV · Paper 02 · extract 1 of 7 · find part b in this mark scheme extract

2 (a)	A1	2.3.4	Set variable base to 50 or heightChk to true	Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements. Don't penalise spelling mistakes and alternative wording of the output.	(10)
	A2	2.2.2	Create WHILE DO loop		
	A3	2.4.1	Request input of height		
	A4	2.2.2	if statement checks if height >= 1 or <= 100		
	A5	2.2.2	if statement checks if height is >= 1 and <=100		
	A6	2.2.2	heightCheck set to false if condition is met		
	A7	2.5.1	Set area using given formula (0.5 x base x height)		
	A8	2.4.1	Display one of base, height and area with appropriate label or displays two of base, height and area without labels		
	A9	2.4.1	Display all of base, height and area with appropriate labels		
	A10	1.1.6	Executing and producing correct output		
Code example					
Python					

```python
#   Q02aFINISHED

#   Initialise variables

base = 50

#   Print prompt and take input from user

    heightChk = TRUE
    while heightChk:
        height = int(input("Enter the height (between 1 and 100): "))
        if (height >= 1 and height <= 100):
            heightChk = FALSE

#   Calculate area and print out values

    area = 0.5 * base * height

    print ("Base length of triangle is : ", base)
    print ("Height of triangle is : ", height)
    print ("Area of triangle is : ", area)
```

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Type MS · Write a short number program from a list of requirements

Answer 3 of 6 · Question 2 part b · 7 marks

2020 NOV · Paper 02 · extract 2 of 7 · find part b in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Write a short number program from a list of requirements

Answer 3 of 6 · Question 2 part b · 7 marks

2020 NOV · Paper 02 · extract 3 of 7 · find part b in this mark scheme extract

2 (b)	B1	2.2.1	Meaningful variable names used (1)	The code example shown is one way of responding to the task. Other methods should be credited accordingly.	(7)
	B2	2.4.1	Meaningful prompts for input (1)		
	B3	2.4.1	Input of length and width as whole numbers (1)		
	B4	2.5.1	Perimeter calculated (= 2 x length + 2 x width) (1)		
	B5	2.5.1	Gap deducted from perimeter (1)		
	B6	2.4.1	Meaningful output (1)		
	B7	2.1.1	Code is fit for purpose. Must have mark points 1-6 and be easy to read (1)		

Code example

Python

```
1   #   Q02bFINISHED
2
3   #   Request input
4
5   length = int(input("Enter the length: "))
6   width = int(input("Enter the width: "))
7
8   #   Calculate number of panels needed
9
10  panels = 2 * length + 2 * width - 4
11
12  #   Print out number of panels needed
13
14  print ("Number of panels needed: " , panels)
```

Type MS · Write a short number program from a list of requirements

Answer 3 of 6 · Question 2 part b · 7 marks

2020 NOV · Paper 02 · extract 4 of 7 · find part b in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Write a short number program from a list of requirements

Answer 3 of 6 · Question 2 part b · 7 marks

2020 NOV · Paper 02 · extract 5 of 7 · find part b in this mark scheme extract

For Q6, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit.

Question	mp	ref	Answer	Additional Guidance	Mark
6	A1	2.2.1	Initialise variable for total sales by member of staff (1)	Sales for each member of staff George Taylor 28466 Fehlix Chayne 35250 Sumren Bergen 30186 Samira Beckle 37430 Nellie Bogart 20448 Cheryl Grouth 34383 Danuta Graunt 26421 Jaiden Deckle 21507 Jimran Caliks 32415 Deynar Derran 28590 ========== Total staff sales: 295096 Highest sales by: Samira Beckle with 37430 Second highest sales by: Fehlix Chayne with 35250 >>>	(11)
	A2	2.5.1	Add succeeding months' sales to total sales (1)		
	A3	2.1.6	Display total sales by member of staff (1)		
	A4	2.2.2	Loop repeats for each member of staff (1)		
	A5	2.2.1	Running team total sales initialised (1)		
	A6	2.2.2	Add succeeding individual staff sales to team total (1)		
	A7	2.1.6	Display final running total (1)		
	A8	2.1.6	Staff member with highest individual sales identified (1)		
	A9	2.1.6	Staff member with second highest individual sales identified (1)		
	A10	2.1.6	Information about staff with highest sales displayed (1)		
	A11	2.1.6	Information about staff with second highest sales displayed (1)		

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Type MS · Write a short number program from a list of requirements

Answer 3 of 6 · Question 2 part b · 7 marks

2020 NOV · Paper 02 · extract 6 of 7 · find part b in this mark scheme extract

Band 1 (1-3 marks)	Band 2 (4-6 marks)	Band 3 (7-9 marks)	Mark
Little attempt to decompose the problem into component parts	Some attempt to decompose the problem into component parts	The problem has been decomposed into component parts	
Some parts of the logic are clear and appropriate to the problem	Most parts of the logic are clear and mostly appropriate to the problem	The logic is clear and appropriate to the problem	
Some appropriate use and manipulation of data types, variables, data structures and program constructs	The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate	The use and manipulation of data types, variables and data structures and program constructs is appropriate	
Parts of the code are clear and readable	Code is mostly clear and readable	Code is clear and readable	
Finished program will not be flexible enough with other data sets or input	Finished program will function with some but not all other data sets or input	Finished program could be used with other data sets or input	
The program meets some of the given requirements	The program meets most of the given requirements	The program fully meets the given requirements	(9)

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Type MS · Write a short number program from a list of requirements

Answer 3 of 6 · Question 2 part b · 7 marks

2020 NOV · Paper 02 · extract 7 of 7 · find part b in this mark scheme extract

Code example

Python

```
allStaffTotal = 0
highestSales = 0
secondSales = 0
high = 0
second = 0

print ("Sales for each member of staff","\n\n")
for staff in staffSales:
    staffTotal = 0
    sales = 3
    while (sales < 9):
        staffTotal = staffTotal + staff[sales]
        sales = sales + 1
    print (staff[1],staff[2],staffTotal)
    if (staffTotal > secondSales):
        if (staffTotal > highestSales):
            highestSales = staffTotal
            high = staff
        else:
            secondSales = staffTotal
            second = staff
    allStaffTotal = allStaffTotal + staffTotal
print ("==========================\n\nTotal sales for all staff: ", allStaffTotal)

print ("\n\nHighest sales by: ",high[1],high[2], " with ", highestSales)
print ("\n\nSecond highest sales by: ",second[1],second[2], " with ", secondSales)
```

Type MS · Write a short number program from a list of requirements

Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 1 of 9 · find part e in this mark scheme extract

1 (a)	A1	**Award 1 mark for any of:** • Comments (1) • Indentation (1) • Meaningful variable/constant/subprogram names (1) • White space (1)		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Write a short number program from a list of requirements

Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 2 of 9 · find part e in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (b)	B1 B2 B3	**Award** 1 mark for each column: **Error description** / **Logic** / **Syntax** / **Runtime** Divide by 0 / / / ✓ Use x instead of * multiply / / ✓ / Subtract 10 from 2 instead of 2 from 10 / ✓ / /		(3)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Write a short number program from a list of requirements

Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 3 of 9 · find part e in this mark scheme extract

1 (c)	C1	**The only correct answer is D** *A is not correct because as it is a data structure* *B is not correct because as there is no arrow looping back to the condition* *C is not correct because as the flowchart does not include an operator*		(1)

Type MS · Write a short number program from a list of requirements

Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 4 of 9 · find part e in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Write a short number program from a list of requirements

Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 5 of 9 · find part e in this mark scheme extract

1 (d)	**Award 1 mark for each of:** D1 = changed to == (1) D2 Condition changed to less than / final two print statements swapped (1) D3 else changed to else (1)		(3)

Code examples

C#

```
if (number1 == number2)
{
    Console.WriteLine("The numbers are equal");
}
else if (number1 < number2)
{
    Console.WriteLine("The highest number is " + number2 + " and the lowest number is " + number1);
}
else
{
```

Java

```
if (number1 == number2)
{
    System.out.print("Numbers are equal");
}
else if (number1 < number2)
{
    System.out.print("The highest number is " + number2 + " and the lowest number is " +number1);
}
else
```

Python

```
if number1 == number2:
    print("Numbers are equal")
elif number1 < number2:
    print("The highest number is", number2, "and the lowest number is", number1)
else:
```

Type MS · Write a short number program from a list of requirements

Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 6 of 9 · find part e in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Write a short number program from a list of requirements

Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 7 of 9 · find part e in this mark scheme extract

1 (e)	**Award 3 marks for:**			
	E1	Prompt for a number to be input (1)		
	E2	Input stored in an integer variable (1)		
	E3	Loop used (1)		
	E4	Loop iterates only for the values 1 – 12 (1)		
	E5	Display is correct when program executes (1)		(5)

Code examples

C#

```
// Print prompt and get number from user
Console.Write("Enter the number: ");
int num = Convert.ToInt32(Console.ReadLine());

// Create loop to display the table
for (int count = 1; count <= 12; count++)
{
    Console.WriteLine(count + " x " + num + " is " + count * num);
}
Console.ReadLine();
```

Java

```
// Print prompt and get number from user
Scanner input = new Scanner(System.in);
System.out.print("Enter the number ");
int num = input.nextInt();

// Create loop to display the table
for (int count = 1; count <=12; count++)
{
    System.out.println(count + " x " + num + " is " + count * num);
}
```

Type MS · Write a short number program from a list of requirements

Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 8 of 9 · find part e in this mark scheme extract

Python	`# Print prompt and get number from user`
	`num = int(input("Enter the number: "))`
	
	`# Create loop to display the table`
	`for count in range(1, 13):`
	`    print(count, "x" , num , "is" , count * num)`

Type MS · Write a short number program from a list of requirements

Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 9 of 9 · find part e in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Write a short number program from a list of requirements

Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 1 of 11 · find part c in this mark scheme extract

Question	Mp	Answer	Additional Guidance	Mark
3 (a) (i)	A1	C# and Python monthNumber/monthsAndDays/count/found (1) Java monthNumber/monthsAndDays/count/found/input		(1)

Type MS · Write a short number program from a list of requirements

Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 2 of 11 · find part c in this mark scheme extract

3 (a) (ii)	A2	monthsAndDays (1)	(1)

Type MS · Write a short number program from a list of requirements

Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 3 of 11 · find part c in this mark scheme extract

3 (a) (iii)	A3	C# • WriteLine, ToString, ToInt32, ReadLine Java • Print, ParseInt, nextLine, ToString, Equals Python • print/len/int/input (1)	(1)

Type MS · Write a short number program from a list of requirements

Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 4 of 11 · find part c in this mark scheme extract

3 (a) (iv)	A4	Display/showMonthNameAndDays (1)	(1)

Type MS · Write a short number program from a list of requirements

Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 5 of 11 · find part c in this mark scheme extract

3 (a) (v)	A5	pMonth (1)	(1)

Type MS · Write a short number program from a list of requirements

Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 6 of 11 · find part c in this mark scheme extract

3 (a) (vi)	A6 A7	**Award 1 mark for each of:** **Type of test data** / **Test data** Boundary (1) / 1 Erroneous (1) / 15		(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Write a short number program from a list of requirements

Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 7 of 11 · find part c in this mark scheme extract

3 (b)	B1 B2	**Award up to 2 marks for a linked explanation such as:** • A library program would not know the names of the calling program's variables in advance (1) meaning it would not be a reusable solution (1) • (Using parameters) enables reusable solutions (1) using different data values/arguments (1) • Using parameters uses less memory / uses memory more efficiently (1) because it avoids the use of global variables / because the memory is freed after the subprogram is finished (1)		(2)

Type MS · Write a short number program from a list of requirements

Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 8 of 11 · find part c in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Write a short number program from a list of requirements

Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 9 of 11 · find part c in this mark scheme extract

3 (c)	**Award 1 mark for each of:**			
	C1	Condition to check whether the year exactly divides by 4 or equivalent condition (1)		
	C2	Condition to check whether year exactly divides by 400 or equivalent condition (1)		
	C3	Condition to check whether year does not exactly divide by 100 or equivalent condition (1)		
	C4	Correct message for at least one of the conditions (1)		
	C5	Program executes and produces the correct output (1)		
	C6	Efficient solution (1)		
				(6)
Code examples				
C#				
Java				

```
if (year % 400 == 0)
{
    Console.WriteLine(year + " is a leap year");
}
else if (year % 4 == 0 && year % 100 !=0)
{
    Console.WriteLine(year + " is a leap year");
}
else
{
    Console.WriteLine(year + " is not a leap year");
}
```

```
if (year % 400 == 0)
{
    System.out.print(year + " is a leap year");
}
else if (year % 4 == 0 && year % 100 !=0)
{
    System.out.print(year + " is a leap year");
}
else
{
    System.out.print(year + " is not a leap year");
}
```

Type MS · Write a short number program from a list of requirements

Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 10 of 11 · find part c in this mark scheme extract

Python	`if year % 400 == 0:`
	`  print(year,"is a leap year")`
	`elif year % 4 == 0 and year % 100 !=0:`
	`  print(year,"is a lear year")`
	`else:`
	`  print(year,"is not a leap year")`

Type MS · Write a short number program from a list of requirements

Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 11 of 11 · find part c in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Write a short number program from a list of requirements

Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 1 of 13 · find part b in this mark scheme extract

3(a)	A1 A2 A3 A4	Award **one** mark for correct column (A OR B) Award **one** mark for correct column (NOT B) Award **one** mark for first set of two cells in third column Award **one** mark for second set of two cells in third column	Award follow through on last column only 5th column can be judged • On columns 1 and 2 for MP3 and MP4 or • as FT from columns 3 and 4 for MP3 and MP4 • award highest mark	4

A	B	A OR B	NOT B	(A OR B) AND (NOT B)
0	0	0	1	0
0	1	1	0	0
1	0	1	1	1
1	1	1	0	0

Type MS · Write a short number program from a list of requirements

Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 2 of 13 · find part b in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Write a short number program from a list of requirements

Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 3 of 13 · find part b in this mark scheme extract

3(b)		Award **one** mark for each of:		
	B1	Correct output taken from array (1)	Must be from array i.e. Indexing used	
	B2	Use of iteration to loop over every item in the array (1)	Award foreach loops Accept any array	
	B3	Use of appropriate method to step accurately, i.e. every other item starting at position 0 (1)	Any appropriate step method Not hard coded index	**3**

Type MS · Write a short number program from a list of requirements

Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 4 of 13 · find part b in this mark scheme extract

C#

```
// Q03bFINISHED

String[] starNames = {"Alasia", "Beid", "Castor", "Denebola",
                      "Electra", "Fafnir", "Gudja", "Haedus",
                        "Izar", "Jishui", "Kang", "Lich",
                        "Maia", "Nahn", "Ogma", "Peacock"};


// Write your code here
int start = 0;
int stop = starNames.Length;
int step = 2;

for (int index = start; index < stop; index = index + step)
{
    Console.WriteLine (starNames[index]);
}

// Alternative
for (int i=0; i<starNames.Length; i=i+2)
{
    Console.WriteLine (starNames[i]);
}

```

Type MS · Write a short number program from a list of requirements

Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 5 of 13 · find part b in this mark scheme extract

Java

```java
// Q03bFINISHED


public class Q03bFINISHED
{
    public static void main (String[] args)
    {
        String starNames[] = {"Alasia", "Beid", "Castor", "Denebola",
                              "Electra", "Fafnir", "Gudja", "Haedus",
                              "Izar", "Jishui", "Kang", "Lich",
                              "Maia", "Nahn", "Ogma", "Peacock"};

        // Write your code here
        int start = 0;
        int stop = starNames.length;
        int step = 2;

        for (int index = start; index < stop; index = index + step)
        {
            System.out.println (starNames[index]);
        }

        // Alternative
        for (int i=0; i<starNames.length; i=i+2)
        {
            System.out.println (starNames[i]);
        }

    }
}
```

Type MS · Write a short number program from a list of requirements

Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 6 of 13 · find part b in this mark scheme extract

Python

```python
# Q03bFINISHED

starNames = ["Alasia", "Beid", "Castor", "Denebola",
             "Electra", "Fafnir", "Gudja", "Haedus",
             "Izar", "Jishui", "Kang", "Lich",
             "Maia", "Nahn", "Ogma", "Peacock"]

# Write your code here

start = 0
stop = len (starNames)
step = 2

for index in range (start, stop, step):
    print (starNames[index])

# Alternative
for index in range (0, len (starNames), 2):
    print (starNames[index])
```

Type MS · Write a short number program from a list of requirements

Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 7 of 13 · find part b in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Write a short number program from a list of requirements

Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 8 of 13 · find part b in this mark scheme extract

3(c)		Award **one** mark for each:		
	C1	Take a string input (1)	• `aString = Console.ReadLine ();` • `aString = myKeyboard.nextLine();` • `aString = input ("Enter a number: ")`	**6**
	C2	Complete the selection with 'if' (1)		
	C3	Convert string to integer (1)	• `aNum = Convert.ToInt32 (aString);` • `aNum = Integer.parseInt (aString);` • `aNum = int (aString)`	
	C4	Any one correct set of relational operators (1) ignore logical operator first two tests ignore test dealing with >= 1	Python • `(aNum <= 20) or (aNum >= 60)` • `(aNum < 21) or (aNum > 59)` • `(aNum > 30) and (aNum < 40)` • `(aNum >= 31) and (aNum <= 39)`	
	C5	Both logical operators used correctly in each test (1) Ignore relational operators First test: **or** (must attempt to address 20 and 60 with relational, even if C4 not awarded Second test: **and**	C# and Java • `((aNum <= 20) || (aNum >= 60))` • `(aNum < 21) || (aNum > 59))` • `((aNum > 30) && (aNum < 40))` • `((aNum >= 31) && (aNum <= 39))`	
	C6	Complete the third test (1)	All Languages: `(aNum == 30)`	

Type MS · Write a short number program from a list of requirements

Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 9 of 13 · find part b in this mark scheme extract

C#

```
// Q03cFINISHED

String aString = "";
int aNum = 0;

Console.WriteLine ("Enter a number:");

// Complete the line to take a string input
aString = Console.ReadLine ();

// Complete the selection statement
if (aString != "")
{
    // Convert the string to an integer
    aNum = Convert.ToInt32 (aString);
    if (aNum > 0)
    {
        // Complete the test
        if ((aNum <= 20) || (aNum >= 60))
        {
            Console.WriteLine ("Acceptable");
        }
        // Complete the test
        else if ((aNum >30) && (aNum < 40))
        {
            Console.WriteLine ("Centre");
        }
        // Complete the test
        else if (aNum == 30)
        {
            Console.WriteLine ("Perfect");
        }
        else
        {
            Console.WriteLine ("No message");
        }
    }
    else
    {
        Console.WriteLine ("The number must be greater than zero");
    }
}
else
{
```

Type MS · Write a short number program from a list of requirements

Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 10 of 13 · find part b in this mark scheme extract

Type MS · Write a short number program from a list of requirements

Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 11 of 13 · find part b in this mark scheme extract

Java

```java
// Q03cFINISHED

import java.util.Scanner;

public class Q03cFINISHED
{
    public static void main (String[] args)
    {
        String aString = "";
        int aNum = 0;
        Scanner myKeyboard = new Scanner (System.in);

        System.out.println ("Enter a number:");

        // Complete the line to take a string input
        aString = myKeyboard.nextLine();

        // Complete the selection statement
        if (aString != "")
        {
            // Convert the string to an integer
            aNum =  Integer.parseInt (aString);
            if (aNum > 0)
            {
                // Complete the test
                if ((aNum <= 20) || (aNum >= 60))
                {
                    System.out.println ("Acceptable");
                }
                // Complete the test
                else if ((aNum >30) && (aNum < 40))
                {
                    System.out.println ("Centre");
                }
                // Complete the test
                else if (aNum == 30)
                {
                    System.out.println ("Perfect");
                }
                else
                {
                    System.out.println ("No message");
                }
            }
            else
            {
                System.out.println ("The number must be greater than zero");
            }
        }
```

Type MS · Write a short number program from a list of requirements

Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 12 of 13 · find part b in this mark scheme extract

Python

```python
# Q03cFINISHED

# Complete the line to take a string input
aString = input ("Enter a number: ")

# Complete the selection statement
if (aString != ""):

    # Convert the string to an integer
    aNum = int (aString)

    if (aNum > 0):

        # Complete the test
        if (aNum <= 20) or (aNum >= 60):
            print ("Acceptable")
        # Complete the test
        elif (aNum > 30) and (aNum < 40):
            print ("Centre")
        # Complete the test
        elif (aNum == 30):
            print ("Perfect")
        else:
            print ("No message")
    else:
        print ("The number must be greater than zero")
else:
    print ("You must provide a number")
```

Type MS · Write a short number program from a list of requirements

Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 13 of 13 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark